

Linux Access Control and Authorization

1. Project description

In this project, I reviewed and modified Linux file and directory permissions to ensure that access was granted only to authorized users. Using Linux permissions management commands, I identified excessive privileges, corrected file access rights, secured hidden files and restricted directory access according to organizational security requirements. This activity demonstrated the practical application of access control and authorization principles.

A) Check file and directory details

Command : `ls -la`

Explanation : I used `ls -la` command to display all files and directories, including hidden files, together with their permission settings. This allowed me to review current access rights before making any modifications.

B) Describe the permissions string

For example : `-rw-rw-r--`

The first character indicates the file type (- means a regular file). The next three characters represent the owner's permissions (rw- = read and write). The following three characters represent group permissions (rw- = read and write). The final three characters represent permissions for others (r-- = read only).

C) Change file permissions.

.Project_k.txt :

User = read, write

Group = read, write

Other = read, write

Then command : `chmod o-w .project_k.txt`

I removed write permissions from the „other” category to align with the organization's security requirements.

D) Change file permissions on a hidden file

.Project_x.txt :

User = read, write

Group = write

Other = none

User and group should be able to read the file. Nobody should have write permissions.

Command :

`chmod u-w .project_x.txt`

`chmod g+r .project_x.txt`

`chmod g-w .project_x.txt`

E) Change directory permissions

Drafts

User = rwx
Group = - - x
Other = - - -

Only researcher2 should access drafts.

Command : `chmod g-x drafts`

User = rwx
Group = - - -
Other = - - -

2. Summary

In this project, I reviewed Linux file and directory permissions and verified whether access controls matched organizational requirements. I used Linux commands to remove excessive permissions, secure a hidden file, and restrict access to a sensitive directory. The project reinforced the principles of authorization, least privileges and access management used in cybersecurity environments.